Theory of Angular Momentum

Ramanuj Raman

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In classical mechanics we cause rotation of a vector $\vec{v}$ by applying a rotation matrix R such that $\vec{v'} = R\vec{v}$ where $\vec{v'}$ is the rotated vector. The rotation matrix is suppose to follow the following properties

- $$R^T R = RR^T = I$$

-

$$\sqrt{v_x^2 + v_y^2 + v_z^2} = \sqrt{v_x'^2 + v_y'^2 + v_z'^2}$$

Now in 3D we can generate all the rotations possible using 3 matricies around x,y and z axis using $R_x(\phi)$, $R_y(\phi)$ and $R_z(\phi)$. For very small angle ϕ, the above matricies allow the commutation relation as $[R_x(\epsilon), R_y(\epsilon)] = R_z(\epsilon^2) - I$

Note 1: Exercise

Generalise the above commuation relation.

Now one of the important lessons of quantum mechanics is that we can rotate a physical system in R^3 then there should be a corresponding operator $\mathcal{D}(R)$ such that it rotates the wavefunction in the hilbert space $\mathcal{H}$ such that $|\alpha\rangle_R = \mathcal{D}(R)|\alpha\rangle$. And the *physics* in the rotated frame is given by $|\alpha\rangle_R$.

Note 2: Exercise

Show that $\mathcal{D}(R)$ is an unitary operator.

Now let us consider a unitary operator which generates the rotation and let $\vec{J} \cdot \hat{n}$ be the generator of the rotation where $\vec{J} = \{\hat{J}_x, \hat{J}_y, \hat{J}_z\}$ and $\hat{n}$ is a unit vector in the direction of the rotation axis. So we can write the $\mathcal{D}(R) = e^{-i\vec{J}\cdot\hat{n}\theta}$. Now in the small angle approximation we have $\mathcal{D}(R) = 1 - i\vec{J} \cdot \hat{n}\theta + \mathcal{O}(\theta^2)$

2.1 Finite Rotation in Quantum mechanics

We can re-write the rotation around an axis as compounding of multiple infinitesimal rotation around the axis as

$$\mathcal{D}_z(\phi) = \lim_{N\to\infty} \left(1 - i\hat{J}_z \frac{\phi}{N}\right)^N$$

which is

$$\mathcal{D}_z(\phi) = exp(-i\hat{J}_z\phi) = 1 - \frac{i\hat{J}_z\phi}{\hbar} - \frac{\hat{J}_z^2\phi^2}{2\hbar^2} + \mathcal{O}(\phi^3)$$

Now let us try to figure-out the angular momentum algebra using the rotation operator. Now we assume that rotation operator have same group properties as the classical rotation matrices.

1. Identity: $R \cdot I = I \cdot R = R \implies \mathcal{D}(R) \cdot I = I \cdot \mathcal{D}(R) = \mathcal{D}(R)$
2. Closure: $R_1 R_2 = R_3 \implies \mathcal{D}(R_1)\mathcal{D}(R_2) = \mathcal{D}(R_3)$
3. Inverses: $RR^{-1} = R^{-1}R = I \implies \mathcal{D}(R)\mathcal{D}(R)^{-1} = \mathcal{D}(R)^{-1}\mathcal{D}(R) = I$

4. Associativity: $R_1(R_2R_3) = (R_1R_2)R_3 \implies \mathcal{D}(R_1)(\mathcal{D}(R_2)\mathcal{D}(R_3)) = (\mathcal{D}(R_1)\mathcal{D}(R_2))\mathcal{D}(R_3)$

Now using the commutation relation of classical rotaion matrix one can workout the commutation relation of the angular momentum operator.

> **Note 3: Exercise**
>
> Workout the commutation relation of angular momentum operator using rotation operator commutation relation, which follows from the fact that they have same group structure.

So one of the central results of this section is the angular momentum algebra given as

$$[\hat{J}_i, \hat{J}_j] = i\hbar\epsilon_{ijk}\hat{J}_k \tag{1}$$

Equation 1 shows that the angular momentum operators forms an **non-abelian group**, which is different from the translation as the generators p_j form an **abelian group**. Now in principle I only need Equation 1 to give all the properties of angular momentum operator and hence rotation operators. But now the question is how the expectation values change with the rotation operator? Consider the expectation value $\langle\hat{J}_x\rangle$, it is defined as

$$\langle\hat{J}_x\rangle = \langle\alpha_R|\hat{J}_x|\alpha_R\rangle = \langle\alpha|\mathcal{D}_z^\dagger\hat{J}_x\mathcal{D}_z|\alpha\rangle$$

Now the let us see how $\hat{J}_x$ transform under the rotation around z-axis?

$$\begin{aligned}\mathcal{D}_z^\dagger\hat{J}_x\mathcal{D}_z = e^{-i\hat{J}_z\phi}\hat{J}_x e^{i\hat{J}_z\phi} &= 1 + \left(\frac{i\phi}{\hbar}\right)[\hat{J}_z, \hat{J}_x] + \frac{1}{2!}\left(\frac{i\phi}{\hbar}\right)^2[\hat{J}_z, [\hat{J}_z, \hat{J}_x]] + \ldots. \\ &= \hat{J}_x\cos(\phi) - \hat{J}_y\sin(\phi)\end{aligned}$$

Now note that these operator transforms as a vector under rotation.

> **Note 4: Exercise**
>
> Show explicitly that the expectation value of angular momemtum operators along different axis transform as a vector under rotation.

3 Spin-$\frac{1}{2}$ Systems and Finite Rotations

The lowest number,N, of dimensions in which the the angular momentum commutation in Equation 1 are realised is $N = 2$. Now the generators of the rotation for $N = 2$ which satisfy the commutation relation are

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \tag{2}$$

$$\sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \tag{3}$$

$$\sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \tag{4}$$

and to make sure the dimensions are satisfied we'll add and extra $\frac{\hbar}{2}$ in front of all of them and they are generator of rotation for spin-$\frac{1}{2}$ particles. Now like showed in the earlier section that the expectation values transform as vector. Let us consider the transformation of a expectation value of spin-$\frac{1}{2}$ matrices around z-axis as,

$$\begin{pmatrix} \cos(\phi) & -\sin(\phi) & 0 \\ \sin(\phi) & \cos(\phi) & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \langle \hat{S}_x \rangle \\ \langle \hat{S}_y \rangle \\ \langle \hat{S}_z \rangle \end{pmatrix} = \begin{pmatrix} \langle \hat{S}_x \rangle \cos(\phi) - \langle \hat{S}_y \rangle \sin(\phi) \\ \langle \hat{S}_x \rangle \sin(\phi) + \langle \hat{S}_y \rangle \cos(\phi) \\ \langle \hat{S}_z \rangle \end{pmatrix} \tag{5}$$

Now the above can be extended to any angular momemtum operator $\hat{J}_i$ as $\hat{J}'_j = R_{ji}\hat{J}_i$. But how a general ket $|\alpha\rangle$ is transformed under rotation,

$$\begin{aligned} e^{-i\hat{J}_z\phi/\hbar}|\alpha\rangle &= e^{-i\hat{J}_z\phi/\hbar}(|0\rangle\langle 0|\alpha\rangle + |1\rangle\langle 1|\alpha\rangle) \\ &= (e^{-i\phi/2}|0\rangle\langle 0|\alpha\rangle + e^{i\phi/2}|1\rangle\langle 1|\alpha\rangle) \end{aligned}$$

Now for $\phi = 2\pi$, $|\alpha\rangle_{R_z(2\pi)} = -|\alpha\rangle$. This is the most astonishing result that after a 2π rotation one pick up a negative sign and in principle it takes two 2π rotations to return to the original state. But in quantum theory we only measure the expectation value of operators and the phase becomes irrevelant so how we detect the sign?

3.1 Pauli Two Component Formalism

Introduced by Wolfgag Pauli in 1927, the two component formalism is a way to represent spin-$\frac{1}{2}$ particles using two component spinors, which becomes more apparent when we consider the full relativistic treatment. As we understood in earlier sections that vectors can be represented using column matrix depending upon the dimensions of the hilbert space. So for spin-$\frac{1}{2}$ particle state can be represented as

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

and

$$|1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

and any vector can be written as

$$|\alpha\rangle = |0\rangle\langle 0|\alpha\rangle + |1\rangle\langle 1|\alpha\rangle = \begin{pmatrix} c_0 \\ c_1 \end{pmatrix}$$

and

$$\langle\alpha| = \langle\alpha|0\rangle\langle 0| + \langle\alpha|1\rangle\langle 1| = \begin{pmatrix} c_0^* & c_1^* \end{pmatrix}$$

where $c_0 = \langle 0|\alpha\rangle$ and $c_1 = \langle 1|\alpha\rangle$. Now, we can represent the pauli matrices using two component formalism as

$$\sigma_x = |0\rangle\langle 1| + |1\rangle\langle 0| = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

$$\sigma_y = -i|0\rangle\langle 1| + i|1\rangle\langle 0| = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$

$$\sigma_z = |0\rangle\langle 0| - |1\rangle\langle 1| = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

Some Properties of Pauli Matrices

1. Hermitian: $\sigma_i^\dagger = \sigma_i$
2. Unitary: $\sigma_i^\dagger = \sigma_i$
3. Traceless: $Tr(\sigma_i) = 0$
4. Determinant: $det(\sigma_i) = -1$
5. Commutation Relation: $[\sigma_i, \sigma_j] = 2i\epsilon_{ijk}\sigma_k$
6. Anticommutation Relation: $\{\sigma_i, \sigma_j\} = 2\delta_{ij}I$
7. Squared: $\sigma_i^2 = I$

Note 5: Exercise

Prove the above properties of pauli matrices.

Now consider $\vec{a} \cdot \vec{\sigma}$ where $\vec{a}$ is a vector and $\vec{\sigma} = (\sigma_x, \sigma_y, \sigma_z)$. We can show that

$$\vec{a} \cdot \vec{\sigma} = a_k \sigma_k = \begin{pmatrix} a_z & a_x - i a_y \\ a_x + i a_y & -a_z \end{pmatrix}$$

Now we'll show a very important identity which will be much useful later,

Note 6: Theorem

$$(\vec{\sigma} \cdot \vec{a})(\vec{\sigma} \cdot \vec{b}) = \vec{a} \cdot \vec{b} I + i \vec{\sigma} \cdot (\vec{a} \times \vec{b})$$

Proof: Let us re-write the left hand side, as $(\vec{\sigma} \cdot \vec{a})(\vec{\sigma} \cdot \vec{b}) = a_i b_j \sigma_i \sigma_j$ and using the commutation and anticommutation relation and

$$\sigma_i \sigma_j = \frac{1}{2}[\sigma_i, \sigma_j] + \frac{1}{2}\{\sigma_i, \sigma_j\} = i \epsilon_{ijk} \sigma_k + \delta_{ij} I \tag{6}$$

hence the left hand side becomes

$$a_i b_j \sigma_i \sigma_j = a_i b_j (i \epsilon_{ijk} \sigma_k + \delta_{ij} I) = \vec{\sigma} \cdot (\vec{a} \times \vec{b}) + \vec{a} \cdot \vec{b}$$

Hence proved.

For $\vec{a} \in \mathbb{R}^3$ we can show that $(\vec{\sigma} \cdot \vec{a})^2 = |\vec{a}|^2 I$ using Equation 6.

3.2 Rotation in Two Component Formalism

As we have shown in the earlier sections that the $\{\sigma_i\}$ are the generators of the rotation in the two dimensional hilbert space. So we can write the rotation operator as a 2×2 matrix as

Note 8: Theorem

$$\mathcal{D}(\hat{n}, \phi) = e^{-i\frac{\phi}{2}\hat{n}\cdot\vec{\sigma}} = \cos\left(\frac{\phi}{2}\right) I - i \sin\left(\frac{\phi}{2}\right)(\hat{n} \cdot \vec{\sigma}) \tag{7}$$

Proof: Using the Taylor expansion of exponential operator we can write

$$e^{-i\frac{\phi}{2}\hat{n}\cdot\vec{\sigma}} = \sum_{k=0}^{\infty} \frac{1}{k!} \left(-i\frac{\phi}{2}\hat{n} \cdot \vec{\sigma}\right)^k$$

Now using the property that for $\hat{n} \in \mathbb{R}^3$, $(\hat{n}\cdot\vec{\sigma})^2 = I$ we can split the above summation into odd and even terms as

$$\begin{aligned} e^{-i\frac{\phi}{2}\hat{n}\cdot\vec{\sigma}} &= \sum_{m=0}^{\infty} \frac{1}{(2m)!}\left(-i\frac{\phi}{2}\right)^{2m} (\hat{n}\cdot\vec{\sigma})^{2m} + \sum_{m=0}^{\infty} \frac{1}{(2m+1)!}\left(-i\frac{\phi}{2}\right)^{2m+1} (\hat{n}\cdot\vec{\sigma})^{2m+1} \\ &= \sum_{m=0}^{\infty} \frac{1}{(2m)!}\left(-i\frac{\phi}{2}\right)^{2m} I + \sum_{m=0}^{\infty} \frac{1}{(2m+1)!}\left(-i\frac{\phi}{2}\right)^{2m+1} (\hat{n}\cdot\vec{\sigma}) \end{aligned}$$

Now using the Taylor expansion of cosine and sine function we get

$$e^{-i\frac{\phi}{2}\hat{n}\cdot\vec{\sigma}} = \cos\left(\frac{\phi}{2}\right) I - i\sin\left(\frac{\phi}{2}\right)(\hat{n}\cdot\vec{\sigma})$$

hence proved.

Now in principle χ which is a 2-component spinor and transforms under rotation as $\chi \to \mathcal{D}(\phi, \hat{n})\chi$ and as I have asked to show that the expectation value of an angular momentum operator transforms as a vector under rotation, and one can go one step ahead and make claims that the angular momentum operator indeed transforms as vector under rotation as $\hat{J}_i' = R_{ij}\hat{J}_j$, but in reality it is $\chi^\dagger\sigma_k\chi$ that transforms as a vector under rotation

$$\chi^\dagger\sigma_i\chi \to R_{ij}\chi^\dagger\sigma_j\chi$$

Now as we have seen earlier that under 2π roatation the spinor picks up a negative sign but what is the eigenvalue of the rotation operator. So we are interested in finding $\mathcal{D}(\phi, \hat{n})\chi = \chi$.

3.3 Eigenvalues and Eigenvectors of Angular Momentum

Consider a new operator defined as

$$\hat{J}^2 = \hat{J}_x^2 + \hat{J}_y^2 + \hat{J}_z^2 = \hat{J}_i\hat{J}_i \tag{8}$$

Note 8: Claim

$\hat{J}^2$ commutes with all the angular momentum operators $\hat{J}_i$.

$$[\hat{J}^2, \hat{J}_i] = 0$$

Proof:

$$\left[\hat{J}_i\hat{J}_i, \hat{J}_j\right] = \hat{J}_i\left[\hat{J}_i, \hat{J}_j\right] + \left[\hat{J}_i, \hat{J}_j\right]\hat{J}_i = i\hbar\epsilon_{ijk}\hat{J}_i\hat{J}_k + i\hbar\epsilon_{ijk}\hat{J}_k\hat{J}_i$$

Now using the property of Levi-Civita symbol $\epsilon_{ijk} = -\epsilon_{jik}$ we can re-write the first term as

$$i\hbar\epsilon_{ijk}\hat{J}_i\hat{J}_k + i\hbar\epsilon_{ijk}\hat{J}_k\hat{J}_i = -i\hbar\epsilon_{jik}\hat{J}_i\hat{J}_k + i\hbar\epsilon_{jki}\hat{J}_k\hat{J}_i$$

Now changing the dummy indices in the first term as $i \to k$ and $k \to i$ we get

$$i\hbar\epsilon_{ijk}\hat{J}_i\hat{J}_k + i\hbar\epsilon_{ijk}\hat{J}_k\hat{J}_i = i\hbar\epsilon_{jki}(-\hat{J}_k\hat{J}_i + \hat{J}_k\hat{J}_i) = 0$$

hence we showed that angular momentum squared operator commutes with all the angular momentum operators. The above proof is valid for system with dimensionality N where $N \geq 2$.

The above claim is very important as it shows that we can find a common eigenbasis for $\hat{J}^2$ and $\hat{J}_i$. Now we can choose the z-axis as the quantization axis and we can find a common eigenbasis for $\hat{J}^2$ and $\hat{J}_z$ as $|j, m\rangle$ where $\hat{J}^2|j, m\rangle \sim \#_1|j, m\rangle$ and $\hat{J}_z|j, m\rangle \sim \#_2|j, m\rangle$ where $\#_1$ and $\#_2$ are the eigenvalues and in few steps will show that $\#_1$ and $\#_2$ are related to j amd m. Now before moving forward let us define the **ladder operators** as

$$\hat{J}_\pm = \hat{J}_x \pm i\hat{J}_y \tag{9}$$

Now consider $\hat{J}_z\hat{J}_\pm|j, m\rangle = (\left[\hat{J}_z, \hat{J}_\pm\right] + \hat{J}_\pm\hat{J}_z)|j, m\rangle = (\#_2 \pm \hbar)\hat{J}_\pm|j, m\rangle$, what this means is that the ladder operator acting on the eigenket $|j, m\rangle$ raises or lowers the eigenvalue of $\hat{J}_z$ by $\hbar$, but it is still an eigenket of $\hat{J}_z$, hence explains the name ladder operator. Now note that $\hat{J}^2\hat{J}_\pm|j, m\rangle = \hat{J}_\pm\hat{J}^2|j, m\rangle = \#_1\hat{J}_\pm|j, m\rangle$, which means that the ladder operator does not change the eigenvalue of $\hat{J}^2$.

Structure of Ladder-Type Operators

There is a generic structure to *ladder type operators*, consider the example of an infinite dimensional hilbert space for a particle in 1D, then we have $\hat{x}$ is the postion operator and $\hat{p}$ is the momentum operator and $[\hat{x}, \hat{p}] = i\hbar$ and we define the translation operator as $\hat{T}(\epsilon) = e^{-i\epsilon\hat{p}/\hbar}$ and $e^{-i\epsilon\hat{p}/\hbar}|x\rangle = \epsilon^n\frac{\mathrm{d}^n}{\mathrm{d}x^n}|x\rangle = |x + \epsilon\rangle$. So here the translation operator acts as a ladder operator on the eigenket of the position operator. But $\hat{x}|x + \epsilon\rangle = (x + \epsilon)|x + \epsilon\rangle$ and the commutation relation $\left[\hat{x}, e^{-i\epsilon\hat{p}/\hbar}\right] = \epsilon e^{-i\epsilon\hat{p}/\hbar}$, using the trick, $[\hat{x}, f(\hat{p})] = i\hbar\frac{\partial f(\hat{p})}{\partial\hat{p}}$. Now let us get the same result using the logic of ladder operators, $\hat{x}\hat{T}(\epsilon) = (x + \epsilon)|x + \epsilon\rangle$ and $\hat{T}(\epsilon)\hat{x}|x\rangle = x|x + \epsilon\rangle$, hence $\left[\hat{x}, \hat{T}(\epsilon)\right] = \epsilon\hat{T}(\epsilon)$.

Now consider the example of a simple harmonic oscillator, we have the ladder operators $\hat{a}$ and $\hat{a}^\dagger$ and the number operator $\hat{N} = \hat{a}^\dagger\hat{a}$ and $\left[\hat{N}, \hat{a}\right] = -\hat{a}$ and $\left[\hat{N}, \hat{a}^\dagger\right] = \hat{a}^\dagger$. Now let us try to generalise the above in terms of a conjecture, consider an operator $\hat{O}$ and $\hat{O}_\pm$ as the corresponding ladder operators. Then $\left[\hat{O}, \hat{O}_\pm\right] = \pm\hat{O}_\pm$. Can we extract more information here? Not sure but for sure there is a generic structure here which can exploieted to solve problems.

Now let us consider the problem of finding the eigenvalues of $\hat{J}_z$ and $\hat{J}^2$ and re-label the eigenkets as $|a, b\rangle$ where $\hat{J}^2 |a, b\rangle = a |a, b\rangle$ and $\hat{J}_z |a, b\rangle = b |a, b\rangle$, now we can write the ladder operator, and $\hat{J}_\pm = c_\pm |a, b \pm \hbar\rangle$, now here goal is to find relation between $c_\pm$and the eigenvalues a and b. Now as shown by the action of $\hat{J}_\pm$ there can't be a indefinite lowering or raising of the eigenvalue of $\hat{J}_z$, while keeping the eigenvalue of $\hat{J}^2$ constant, as $\hat{J}^2 = \sum_i \hat{J}_i^2$ hence $a \geq b^2$, now exploiting this assertion we have

$$\hat{J}^2 = \hat{J}_x^2 + \hat{J}_y^2 + \hat{J}_z^2 = \frac{1}{2}(\hat{J}_+\hat{J}_- + \hat{J}_-\hat{J}_+) + \hat{J}_z^2$$

this implies

$$\hat{J}^2 - \hat{J}_z^2 = \frac{1}{2}(\hat{J}_+\hat{J}_- + \hat{J}_-\hat{J}_+)$$

and the RHS is a positive semi-definite operator, by normalization condition, hence asseration $a \geq b^2$ is satisfied. Now let us consider the maximum value of b as b_{max}, then $\hat{J}_+ |a, b_{max}\rangle = 0$, and using the above relation we have

$$\hat{J}_-\hat{J}_+ |a, b_{max}\rangle = \hat{J}_x^2 + \hat{J}_y^2 + i\left[\hat{J}_x, \hat{J}_y\right] = (\hat{J}^2 - \hat{J}_z^2 - \hbar\hat{J}_z) |a, b_{max}\rangle = 0$$

$$a - b_{max}^2 - \hbar b_{max} = 0 \implies a = b_{max}(b_{max} + \hbar)$$

Similarly for $\hat{J}_+\hat{J}_- |a, b_{min}\rangle = 0$, we have $a = b_{min}(b_{min} - \hbar)$ now these two conditions will give us the relation $b_{max} = -b_{min}$ and $b_{max} \geq b \geq -b_{max}$ and after some finite actions of $\hat{J}_+$ we'll have $b_{max} = b_{min} + n\hbar$ hence we have $b_{max} = \frac{n\hbar}{2}$ and let $j = \frac{n}{2}$ and hence we have $b_{max} = j\hbar$ and $a = \hbar^2 j(j+1)$ and let us define $b \equiv m\hbar$ and hence we have $m \in \{-j, -j+1, ..., j-1, j\}$ where cardinality is $2j+1$ and the eigenvalues of $\hat{J}^2$ and $\hat{J}_z$ are given as

$$\hat{J}^2 |j, m\rangle = \hbar^2 j(j+1) |j, m\rangle \tag{10}$$

$$\hat{J}_z |j, m\rangle = \hbar m |j, m\rangle \tag{11}$$

So if j is an integer then m is also an integer and if j is a half-integer then m is also a half-integer.

Dimensionality of Hilbert Space

The cardinality is also the dimension of the hilbert space and $j = \frac{N-1}{2}$ where N is the dimension of the hilber space. Note that here the dimension of the hilbert space have nothing to do with the dimension of the physical space, as we have seen that for spin-$\frac{1}{2}$ particle the hilbert space is 2 dimensional but the physical space can be determined by the experimental setting.

4 Matrix Representation of Operators

4.1 Angular Momentum Operators

Now we have the eigenvalues of $\hat{J}^2$ and $\hat{J}_z$, we can write the matrix representation of the angular momentum operators in the eigenbasis of $\hat{J}^2$ and $\hat{J}_z$ as

$$\langle j', m'|\hat{J}^2|j, m\rangle = \hbar^2 j(j+1)\delta_{jj'}\delta_{mm'} \tag{12}$$

$$\langle j', m'|\hat{J}_z|j, m\rangle = \hbar m\delta_{jj'}\delta_{mm'} \tag{13}$$

Now more important is the matrix representation of the ladder operators, which can be derived using the following relation

$$\hat{J}_\pm |j, m\rangle = c_\pm |j, m \pm 1\rangle$$

$$\langle j, m| \hat{J}_\mp \hat{J}_\pm |j, m\rangle = |c_\pm|^2$$

Now lhs is $\hat{J}_\mp \hat{J}_\pm = \hat{J}^2 - \hat{J}_z^2 \pm i\left[\hat{J}_x, \hat{J}_y\right]$,hence we have $c_\pm = \hbar\sqrt{j(j+1) - m(m\pm 1)}$ so we have

$$\langle j', m'|\hat{J}_\pm|j, m\rangle = \hbar\sqrt{j(j+1) - m(m\pm 1)}\delta_{jj'}\delta_{m', m\pm 1} \tag{14}$$

4.2 Rotation Operators

Now consider the rotation operator $\mathcal{D}(R)$, where R is a rotation defined by angle ϕ and axis $\hat{n}$. Hence its matrix elements are given as

$$\mathcal{D}^{(j)}_{m'm}(R) = \langle j, m'|\mathcal{D}(R)|j, m\rangle = \langle j, m'|e^{-i\hat{J}\cdot\hat{n}\phi/\hbar}|j, m\rangle$$

Note that there are no matrix elements between different j's as $\hat{J}^2$ commutes with $\mathcal{D}(R)$ and hence the rotation operator is block diagonal in the eigenbasis of $\hat{J}^2$ and $\hat{J}_z$. Now note that the matrix formed by this is the irreducible representation of the rotation operator, and is called the Wigner D-matrix. The dimension of this matrix is $2j + 1$ and hence for spin-$\frac{1}{2}$ particle the dimension of the Wigner D-matrix is 2. Like the Rotation operator, the Wigner D-matrix also satisfies the group properties for each j.

Now we can use α,β,γ as the Euler angles to represent the rotation operator as $\mathcal{D}(R) = \mathcal{D}(\alpha, \beta, \gamma) = e^{-i\hat{J}_z\alpha/\hbar}e^{-i\hat{J}_y\beta/\hbar}e^{-i\hat{J}_z\gamma/\hbar}$ and hence we can write the matrix elements of the Wigner D-matrix as

$$\mathcal{D}^{(j)}_{m'm}(\alpha, \beta, \gamma) = \langle j, m'|e^{-i\hat{J}_z\alpha/\hbar}e^{-i\hat{J}_y\beta/\hbar}e^{-i\hat{J}_z\gamma/\hbar}|j, m\rangle$$

5 Orbital Angular Momentum

So in earlier in section we talked about the angular momentum operator $\hat{J}$ as a generator of infitesimally small rotation. Now earlier we talked about spin angular momentum which is an intrinsic property of the particle, but there is another type of angular momentum which is called orbital angular momentum and it is related to the motion of the particle in space. Now we can define the orbital angular momentum operator, following the same idea in classical mechanics as

$$\hat{L} = \hat{x} \times \hat{p} \tag{15}$$

5.1 Orbital Angular Momentum as generator of Rotations

In this next seciton I'll show that the orbital angular momentum operator is generator of rotation in the position space and hence it satisfies the same commutation relation as the angular momentum operator $\hat{J}$.

Note 9: Orbital Angular Momentum Commutation Relation

The orbital angular momentum operator satisfies the same commutation relation as $\hat{J}$

$$\left[\hat{L}_i, \hat{L}_j\right] = i\hbar\epsilon_{ijk}\hat{L}_k \tag{16}$$

Proof: Consider the angular momentum operator $\hat{L}_i = \epsilon_{iab}\hat{x}_a\hat{p}_b$ and $\hat{L}_j = \epsilon_{jcd}\hat{x}_c\hat{p}_d$, now we can write the commutator as

$$\left[\hat{L}_i, \hat{L}_j\right] = \epsilon_{iab}\epsilon_{jcd}[\hat{x}_a\hat{p}_b, \hat{x}_c\hat{p}_d]$$

Now using the identity $[AB, CD] = A[B, C]D + [A, C]DB + C[A, D]B + CA[B, D]$ and the canonical commutation relation $\left[\hat{x}_i, \hat{p}_j\right] = i\hbar\delta_{ij}$ we can write the above as

$$\left[\hat{L}_i, \hat{L}_j\right] = i\hbar\epsilon_{iab}\epsilon_{jcd}(\delta_{bc}\hat{x}_a\hat{p}_d - \delta_{ad}\hat{x}_c\hat{p}_b) = i\hbar(\epsilon_{iab}\epsilon_{jcb}\hat{x}_a\hat{p}_d - \epsilon_{iab}\epsilon_{jcd}\delta_{ad}\hat{x}_c\hat{p}_b)$$

Now using the identity $\epsilon_{iab}\epsilon_{jcb} = \delta_{ij}\delta_{ac} - \delta_{ic}\delta_{aj}$ and $\epsilon_{iab}\epsilon_{jcd}\delta_{ad} = \delta_{ij}\delta_{bc} - \delta_{ic}\delta_{bj}$ we can write

$$\left[\hat{L}_i, \hat{L}_j\right] = i\hbar(\delta_{ij}\hat{x}_a\hat{p}_a - \delta_{ic}\hat{x}_j\hat{p}_c - \delta_{ij}\hat{x}_c\hat{p}_c + \delta_{ic}\hat{x}_c\hat{p}_j) = i\hbar(\hat{x}_i\hat{p}_j - \hat{x}_j\hat{p}_i) = i\hbar\epsilon_{ijk}\hat{L}_k$$

hence we have shown that the orbital angular momentum operator satisfies the same commutation relation as the angular momentum operator $\hat{J}$.

Now to put the nail in the coffin, we show that $e^{-i\delta\phi\hat{L}_z/\hbar}$ is indeed a rotation operator in the position space. Consider a state $|x', y', z'\rangle$ and the action of the rotation operator on this state is given as

$$\left(1 - i\frac{\delta\phi}{\hbar}\hat{L}_z\right)|x', y', z'\rangle = \left(1 - i\frac{\delta\phi}{\hbar}(\hat{x}\hat{p}_y - \hat{y}\hat{p}_x)\right)|x', y', z'\rangle$$

Now because $\left[\hat{x}_i, \hat{p}_j\right] = i\hbar\delta_{ij}$, we can swtich the order of the operators above and write the above as

$$\left(1 - i\frac{\delta\phi}{\hbar}(\hat{p}_y\hat{x} - \hat{p}_x\hat{y})\right)|x', y', z'\rangle = \left(1 - i\frac{\delta\phi}{\hbar}(\hat{p}_y x' - \hat{p}_x y')\right)|x', y', z'\rangle$$

Now with here the second term is a translation operator acting in the $x - y$ plane and we can write the above as

$$\left(1 - i\frac{\delta\phi}{\hbar}(\hat{p}_y x' - \hat{p}_x y')\right)|x', y', z'\rangle = \left(1 - i\frac{\delta\phi}{\hbar}(\hat{p}_y x' - \hat{p}_x y')\right)|x', y', z'\rangle = |x' - y'\delta\phi, y' + x'\delta\phi, z'\rangle$$

hence we have shown that the orbital angular momentum operator is indeed a generator of rotation in the position space. Now it comes with no surpries that we should be using spherical coordinates to represent the orbital angular momentum operator. Consider the wavefunction as $|\alpha\rangle$, then the action of the orbital angular momentum operator on the wavefunction is given as

$$\langle r, \theta, \phi| \left(1 - i\frac{\delta\phi}{\hbar}\hat{L}_z\right)|\alpha\rangle = \langle r, \theta, \phi - \delta\phi|\alpha\rangle = \langle r, \theta, \phi|\alpha\rangle - \delta\phi\frac{\partial}{\partial\phi}\langle r, \theta, \phi|\alpha\rangle$$

hence we can write the orbital angular momentum operator in spherical coordinates as

$$\hat{L}_z = -i\hbar\frac{\partial}{\partial\phi}$$